

Statistical Learning with High-dimensional Data



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Preamble

"Ce qui est simple est toujours faux.
Ce qui ne l'est pas est inutilisable."

Paul Valéry

Outline

1. Introduction
2. Reminder of descriptive statistics
3. The learning process
4. Supervised learning
5. Unsupervised learning
6. Learning in high-dimensions
7. Dimension reduction
8. Clustering in high dimensions

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The AI revolution hasn't happened yet!

Artificial intelligence is a strategic field of research:

- with direct application in most scientific fields (Medicine, Biology, Astrophysics, Humanities)
- and with probably the most impact in innovation and transfer (health, transport, defense).

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The recent and impressive NN results **should not hide the remaining issues**:

- deep learning has impressive results in a few specific cases and with a high-level supervision,
- use of DL techniques in various fields are promising but not well understood.

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- use of DL techniques in various fields are promising but not well understood.

"Artificial Intelligence: the revolution hasn't happened yet"

M. Jordan (UC Berkley)

Open problems of AI

Some open problems are critical:

- reliability of models and algorithms,
- handling data heterogeneity (categorical, functional, networks, images, texts, ...),
- unsupervised learning (clustering, dimension reduction),
- learning from HD and small data (n small / p large),

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Combination of statistical theory with deep learning techniques is certainly the future of AI!

AI in France

French policy for AI: since 2018

- C. Villani presented in March a recommendation report for AI,
- President Macron announced the creation of a network of AI institutes.



The 3IA institutes: (now AI clusters)

- 12 french research centers applied for the 3IA call in Sept.,
- 4 projects have been selected in the Spring 2019:
 - Paris, Toulouse, Grenoble
 - and Nice!

5 additional institutes were added in 2024



A few examples: Cervical cancer detection

Cervical cancer detection:

- it is an important public health field which is currently treated mostly manually,
- screening by human experts is complicated by the amount of cells (20 000/smear),
- and by the very small proportion of cancer cells (less than 1%).



Figure: Normal (left) and abnormal (right) pap smears.

Classification is useful in this context:

- for building supervised classifiers which can select the most likely cancer cells,
- for helping experts in labeling the learning data through weakly-supervised classification,
- for selecting discriminative variables which can be used in a semi-automatic process.

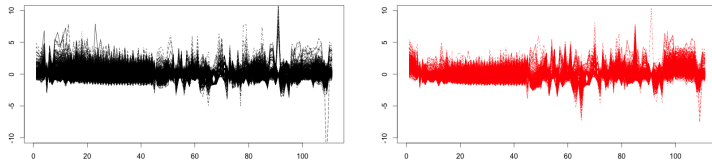


Figure: Control and (cervical) cancer data

A few examples: Sparse models in Medicine (HEGP)

Problem:

- overcome the curse of dimensionality that occurs in Metabolomics,
- for disease diagnostic and early-stage marker identification,
- metabolomic data fall into the "ultra-high dimensional data" case.

Our solution:

- a Bayesian variable selection technique for PCA,
- that identify the relevant variable for each stage of the disease.

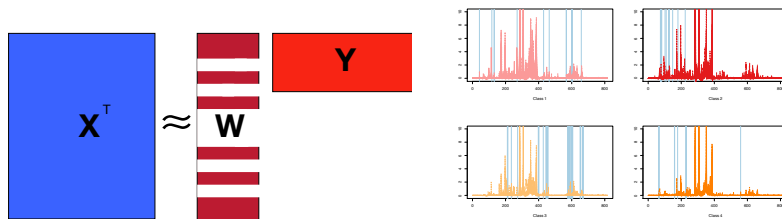


Figure: ~~Functional clustering of Linky data (EDF).~~

Analysis of massive functional data (Linky / EDF)

Problem:

- Linky meters will allow EDF to have access to 27 million of Linky data,
- data are functional data and are measured every 30 minutes -> 17 520 obs./year,
- necessity to summarize those massive data before exploitation.

Our solution:

- a statistical co-clustering technique for functional data,
- that form homogeneous groups of both individuals and days.

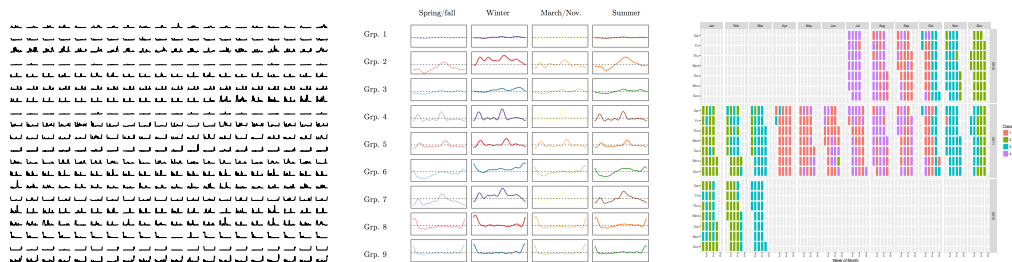


Figure: Functional co-clustering of Linky data (EDF).

Outline of the course

5 first sessions (C. Bouveyron):

- Introduction
- Dimension reduction: PCA, PPCA, FDA
- The GMM and EM algorithms
- Subspace methods for clustering and classification
- Analysis of functional data

5 following sessions (P.-A. Mattei):

- High-dimensional linear regression
- Sparse linear regression and the lasso
- Bayesian linear regression
- Sparsity beyond linear regression 1: classification, recommender systems
- Sparsity beyond linear regression 2: PCA, clustering

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The different types of data:

- continuous data: $x \in \mathbb{R}$ or $\mathbb{R}^p$ where p is the nb of variables.
- discrete data: $x \in \{1, 2, \dots, 10\}$
 - continuous
 - count data
 - ordinal data (ratings)
 - categorical data



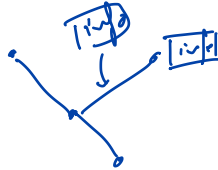
→ $x \in \{1, 2, 3, \dots, 5\}$ ⇒ $\{\text{blue}, \text{red}, \text{pink}\}$
no order 1 ~~X~~ 2
no distance

less frequent types of data:

- images  $x_{ij} \in [0, 255]^3$

- functional data / time series: recent expansion of this type of data thanks to the smart sensors

- network data:



- text data:



probably more info encoded in a text than in a single label.

The general processing of a data study.

1) Qualify and translate the question.

2) Creation of the data set.

3) Pre-treatment of the data (missing data, abnormal data, extreme data).  Could take a long time!

4) Technical part of the analysis
↳ classification, regression, clustering, ...

5) Communication of your results: reports, slides, ...

14
6) Deployment of the solution into the company

The pre-treatment of data :

- the missing data :

⊕ easy to find !

⊖ most of the methods are not accepting missing data.

⇒ the solution to that :

1) hard solution : remove individuals or variables with ^{hard} more than $X\%$ of missing data ($X = 70-80\%$)

2) impute the data : • with the mean/median or value ¹⁵ to the individual. • use a specific algorithm to adapt the

(the EM algorithm)

- abnormal data:

⊖ not easy to identify the data → define "normality" rules for each variable and apply them to the data.

↳ remove the abnormal data and treat them as missing data.

- extreme data:

in some cases, abnormal data are in fact actual data and should be considered as extreme data
⇒ theory of extreme data.

Descriptive statistics

1) Unidimensional data:

(i) quantitative data:

numerical

- numerical summary:

- indicators of the central tendency:
mean, median, modes
- _____ for the dispersion:
std, variance, range, inter-quartile dist.
+ min, max, quartiles, ...

inter quantile distance

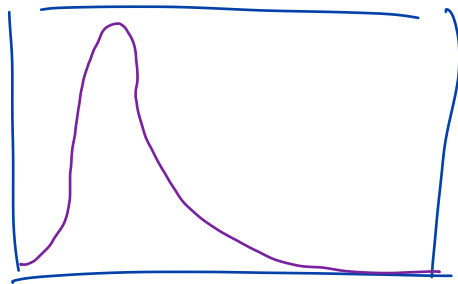
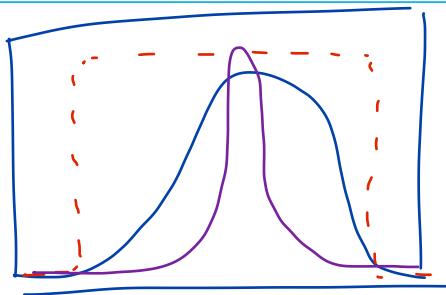
- graphical summary:

- **histogram**: it aims at looking at the density / distribution of the data / variable.

⚠ finding the right nb of slices in the hist. is a difficult task.

- **boxplot**: ^{same aim} same aim as the histogram: look at the distribution of the data.





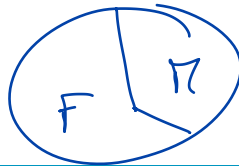
(ii) qualitative data:

- numerical summary: frequencies of the different modalities
- graphical summary:

- bar plot:



- pie plot:



Multivariate data:

(i) quantitative data:

- numerical summary:

- central tendency:

- vector of means : $\underline{m} = (m_1, m_2, \dots, m_p)$

- dispersion:

- variance-covariance matrix : $\underline{P}$

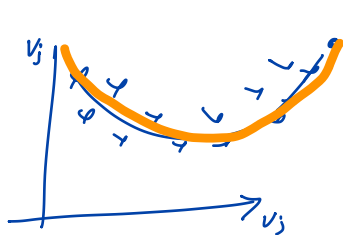
covariance between
 V_i and V_j : σ_{ij}

σ_{ij}

σ_i^2

σ_p^2

The covariance matrix is also symmetric



correlation matrix:



correlation between v_i and v_j

between -1 and 1

$$r_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j} \in \underline{[-1, 1]}$$

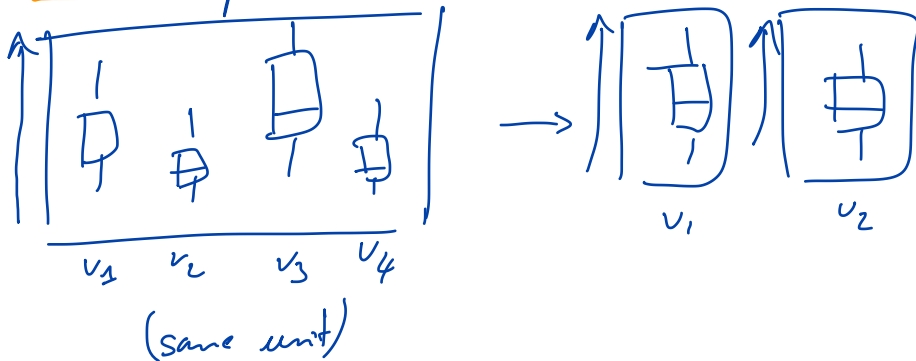
if $r_{ij} \sim 0$: the two variables are not linearly correlated (they possibly carry diff. info)

$r_{ij} \sim 1$: v_i and v_j are correlated = same info

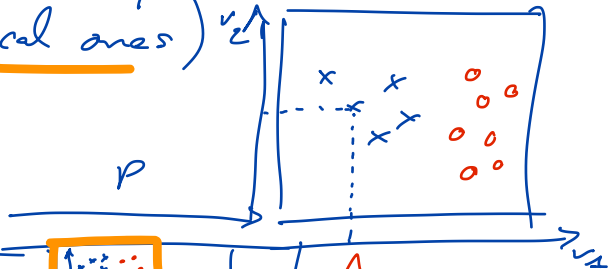
$r_{ij} \sim -1$: v_i and v_j are anti-correlated.

- graphical summary:

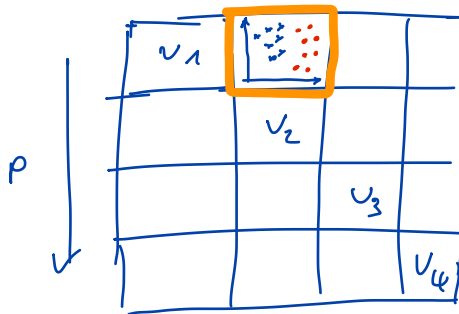
- boxplots for several variables



- scatter plots : for 2 quantitative variables
(+ 1 or 2 categorical ones)

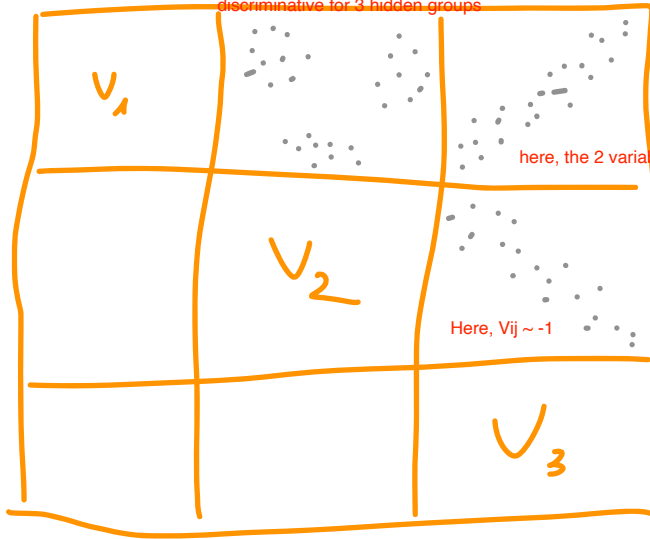


- pair plot:



⚠ interesting tool
but which only
proposes a partial
view of the data.

Here, the two variables are
discriminative for 3 hidden groups



here, the 2 variables are correlated ($V_{ij} \sim 1$)

Here, $V_{ij} \sim -1$

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Learning from data...

One task, several families of approaches:

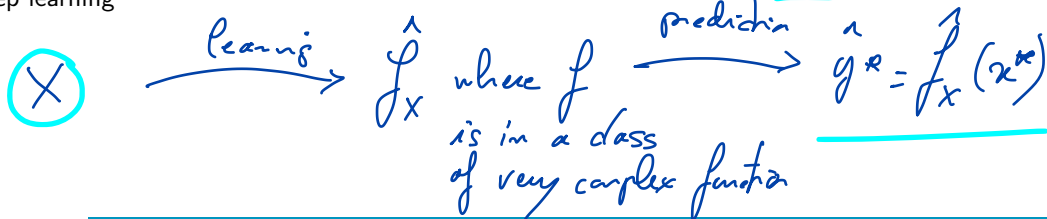
Statistical learning



Machine learning



Deep learning



...

Learning from data...

Learning is a two-head problem:

Supervised

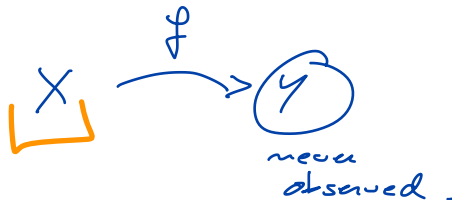
Learning from two sources of data:

- X: explanatory variables
- Y: target variables

and we need examples from the two variables to learn a predictor f .

Unsupervised

in this context we aim at learning only from X to predict a target variable which is ^{never} observed!



Learning from data...

Methods are specific to each task:

Supervised

- regression:
 - Y is a continuous variable
- classification:
 - Y is a categorical var.
- time series forecasting:


Unsupervised

- clustering:
 - Y is categorical
 - dimension reduction / visualization
representation learning:
 - Y is (multivariate) continuous
- $$X \longrightarrow Y \in \mathbb{R}^d$$
- image denoising, ...

Supervised learning

Supervised learning is also a field with different sub-tasks, with specific methods:

- classification:

- random forests
- logistic regression
- k-NN
- SVM
- CNN
- ...

- regression:

- Linear model (OLS)
- Generalized linear model (GLM)
- SVM
- Rand. Forest
- NN

- time series analysis:

- Arma

- LSTM
- XGboost

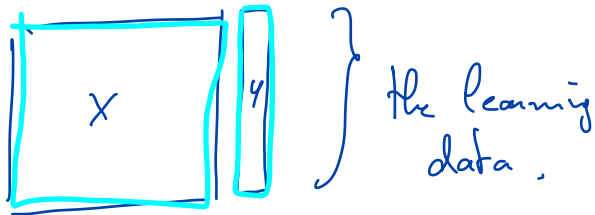
- ...

	SL	ML
Supervised problems	Logistic regression	k-NN SVM
Unsupervised problems	t-SNE	k-means

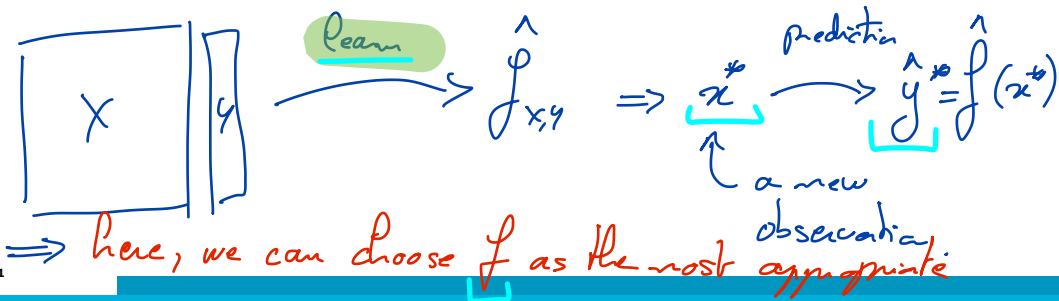
The supervised learning process

The material: a set of (complete) data

$$(x, y)$$



The goal: learn a predictor $f(\cdot)$ from the (complete) data



Measuring the learning performance

One comfortable thing of working in the supervised context is:

- to be able to measure the performance of the learned predictor,

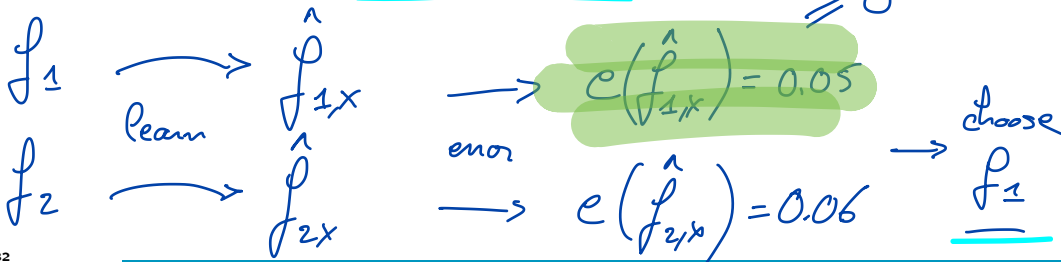
- classification error:

$$e(\hat{f}) = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{\hat{y}_i \neq y_i\}}{\in [0,1]}$$

- regression:

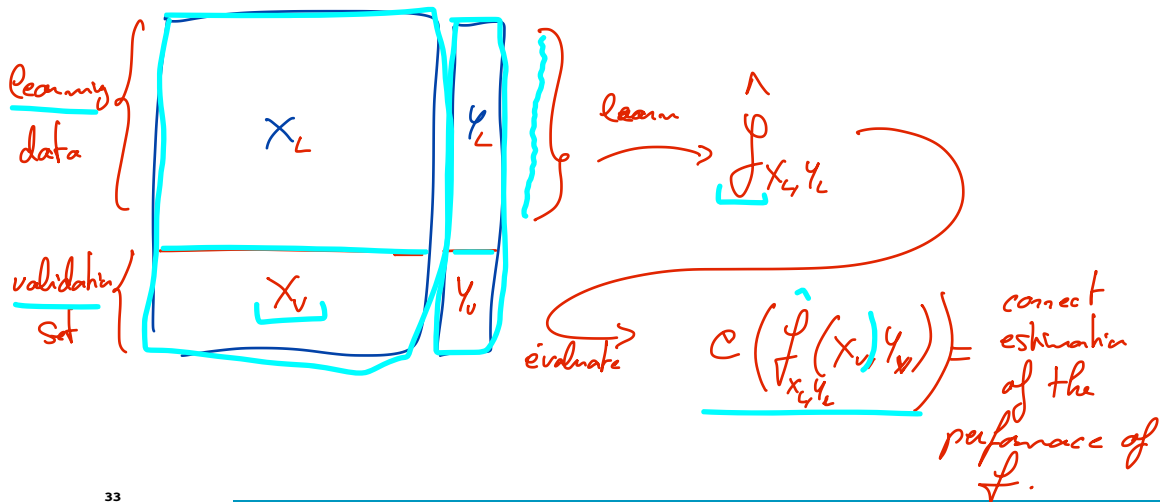
$$MSE(\hat{f}) = \frac{1}{n} \sum_{i=1}^n \|\hat{y}_i - y_i\|^2$$

- compare several predictors and pick the most efficient one.



A minimal setup for supervised learning

The minimal setup for building a supervised predictor $f()$ from data is as follows:



Why such a minimal setup?

The goal is to avoid **over-fitting** when choosing the model or the model parameters:

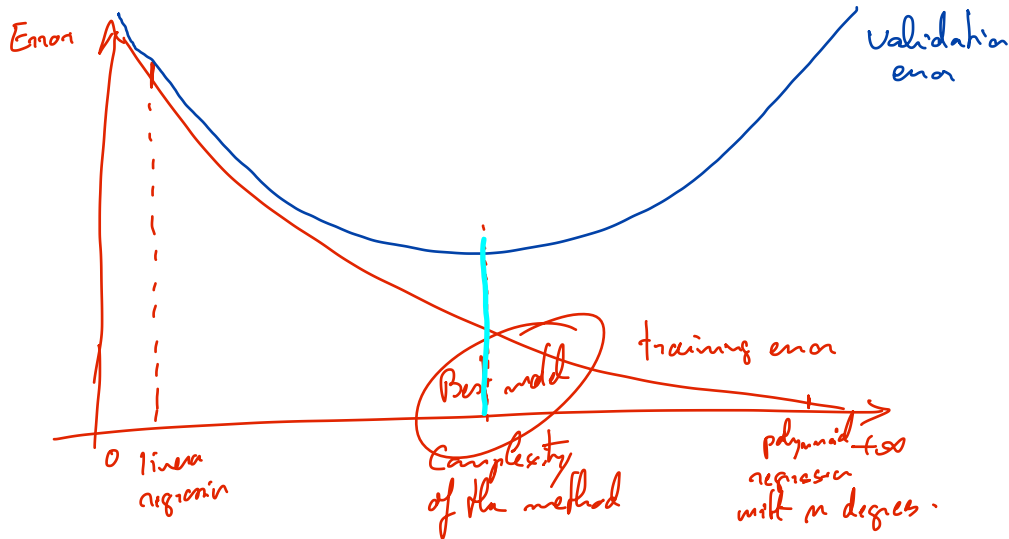
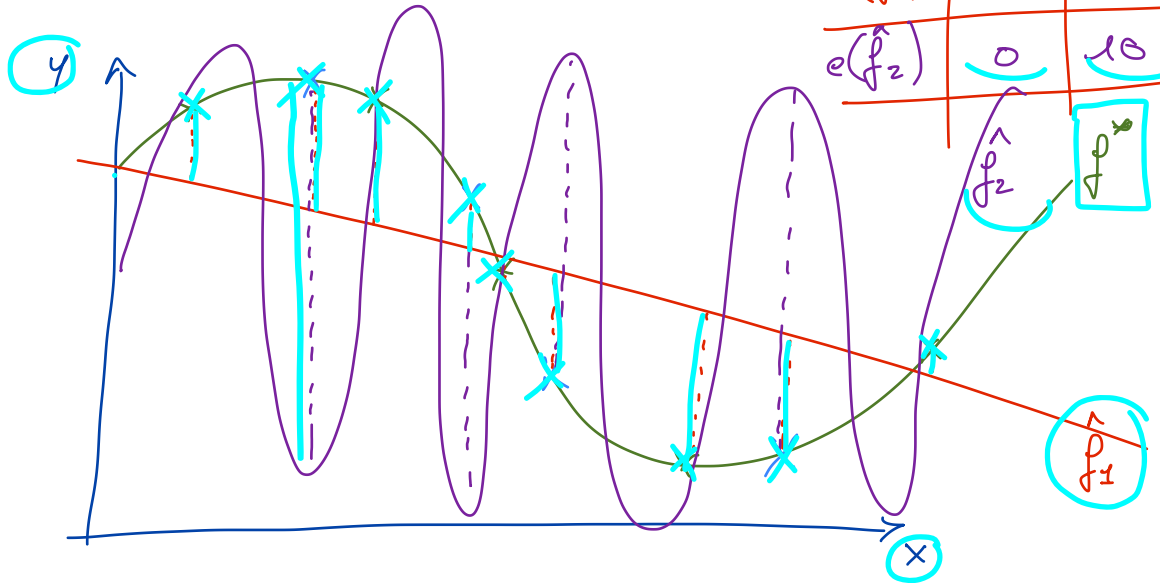


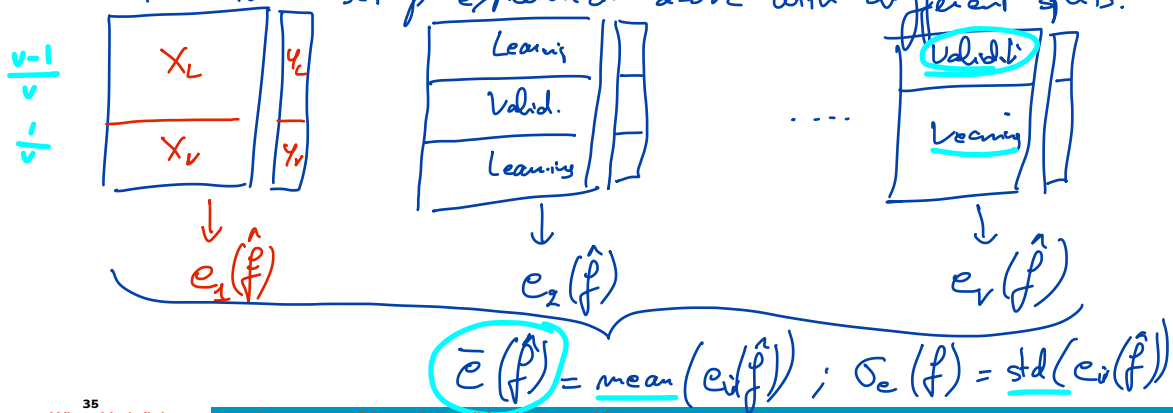
Illustration of overfitting in regression:



An advanced setup for supervised learning

Resampling techniques:

- there are several methods (leave-one-out, V-fold cross-validation, bootstrap) depending on the context (sample size, computing time, ...),
- V-fold cross-validation: The main idea of V-fold CV is to replicate the minimal setup explained above with different splits.



This single approach allows you to have a better comparison of the methods you would like to challenge.

$$f_1 = \text{svm}$$

$$f_2 = \text{log-reg}$$

$$f_3 = \text{h-nn}$$



$$e(\hat{f}_1) = 0.05 \pm 0.02$$

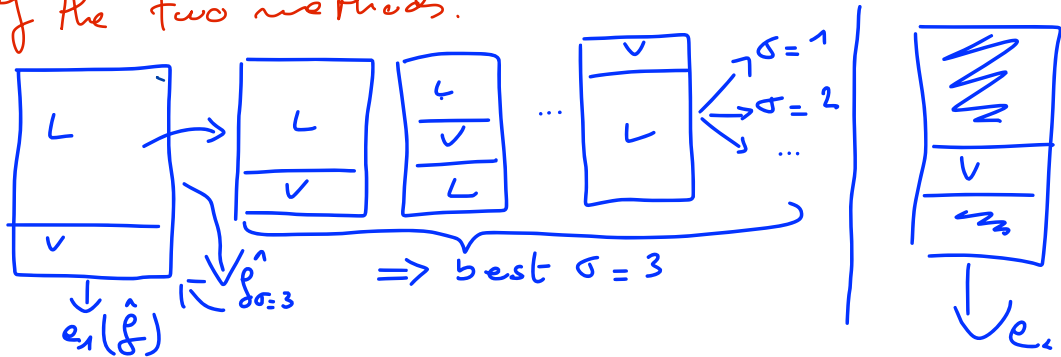
$$e(\hat{f}_2) = 0.06 \pm 0.01$$

$$e(\hat{f}_3) = 0.09 \pm 0.03$$

A final point on this is to notice that most supervised learning techniques have hyperparameters to tune. The best way to select them is to use V -fold CV and consider one method with two values for the hyperparameter as two different methods!

$$\begin{array}{ll} f_1 = \text{SVM Linear} & \longrightarrow e(\hat{f}_1) \\ f_2 = \text{SVM RBF}, \sigma=1 & \longrightarrow e(\hat{f}_2) \\ f_3 = \text{_____} \quad \sigma=2 & \longrightarrow e(\hat{f}_3) \end{array}$$

⚠ Important concept: when comparing one method with no hyperparameter and another with a hyperparameter to tune, we have to use **double cv** to correctly compare the perf. of the two methods.



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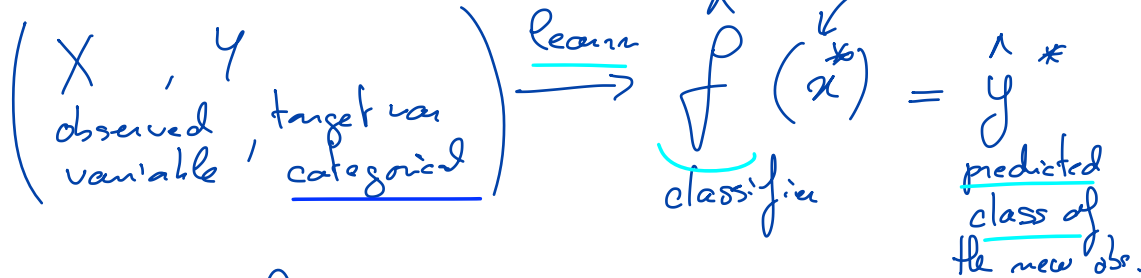
for v in 1:V
    . split the data
    . learn SVR linear ( $f_1$ ) on  $X_L, Y_L$ 
    for v inside 1:V
        split  $X_L, Y_L$ 
        learn SVR Rbf  $\sigma = 1 \dots 10$  ( $f_2$ )
        evaluate the error
    pick the best  $\sigma^*$ 
    evaluate SVR linear and SVR RBF  $\sigma^*$ 
end
boxplot  $e(\hat{f}_1)$  and  $e(\hat{f}_2)$ 

```

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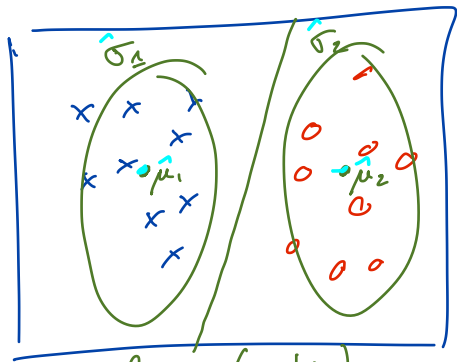
Classification:



Two main families of methods to address classif.:

- (i) Generative methods (SL)
those methods assume a model for each class and deduce the classification rule from this model
- (ii) Discriminative methods (NL)
those methods directly model the classification rule.

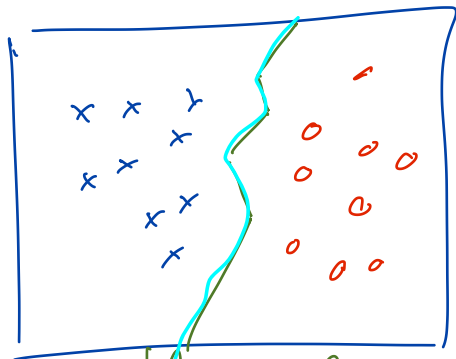
Generative



we model $p(x|y)$
and we deduce $P(y|x)$
afterward

⇒ linear discriminant analysis LDA
⇒ Logistic regression

Discriminative



we directly model
 $P(y|x)$

⇒ SVM
⇒ k-NN
⇒ CNN.

Linear discriminant analysis (LDA or FDA)

Fisher proposed this method in 1936 and was the first classification method ever ^{proposed}. Since then, it ~~has~~ become a reference method (cheap one and performs often well)

The rationale behind LDA is simple: LDA assumes that each class is distributed as a Gaussian distribution with a specific mean and a common covariance matrix.

The model:

$$p(x|y=l) \sim \mathcal{N}(x; \mu_l, \Sigma) \quad \forall l=1 \dots K$$

$$p(y=l) = \pi_l.$$

1) Learn the model (X, Y)

we have to estimate

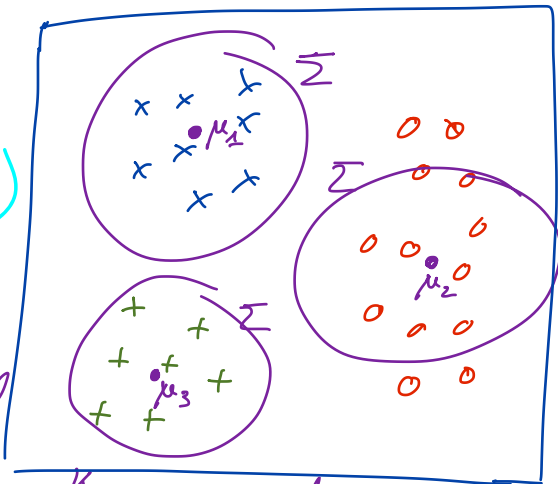
$$\{\mu_1, \mu_2, \mu_3, \Sigma, \pi_1, \pi_2, \pi_3\}$$

using the Maximum Likelihood

$$\hat{\mu}_l = \frac{1}{n_l} \sum_{i=1}^n \mathbb{1}\{y_i=l\} x_i$$

$$\hat{\pi}_l = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{y_i=l\}$$

$$\hat{\Sigma} = \frac{1}{n} \sum_{l=1}^K n_l S_l \quad \text{where} \quad S_l = \frac{1}{n_l} \sum_{i=1}^n \mathbb{1}\{y_i=l\} x_i x_i^T$$



3) Obtaining the classification function:

Thanks to the Bayes' theorem, we can get:

$$\underline{P(Y=k|X)} = \frac{P(Y=k) P(X|Y=k)}{P(X)} = \frac{\hat{\pi}_k N(\hat{x}; \hat{\mu}_k, \hat{\Sigma})}{\sum_{l=1}^K \hat{\pi}_l N(\hat{x}; \hat{\mu}_l, \hat{\Sigma})}$$

when a new value x^* is coming, we can assign this individual to the class the most probable a posteriori: (map rule)

$$\hat{y}^* = \underset{k}{\operatorname{argmax}} P(Y=k|X^*) = \underset{k}{\operatorname{argmax}} \hat{\pi}_k N(\hat{x}^*; \hat{\mu}_k, \hat{\Sigma})$$

$\hat{f}(x^*)$

As a summary on LDA:

⊕ LDA is by nature working with several classes ($K \geq 2$)

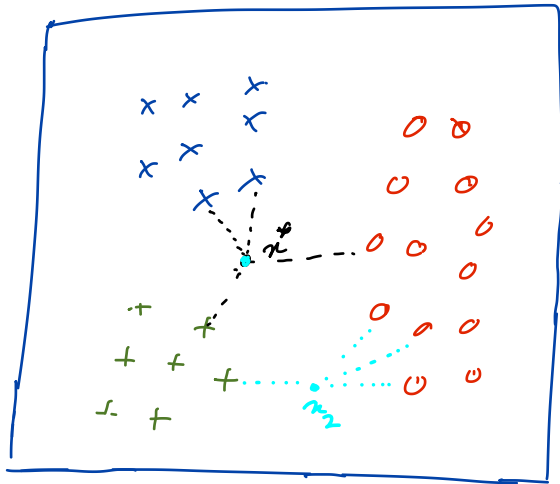
- ⊕ it is a reference technique / old but still working well
- ⊕ the model is easy to understand
- ⊕ ML + Bayes' Theorem allows to get the classification rule
- ⊕ knowledge of the classification probability
- ⊕ the method is quite robust to the possible non-Gaussianity of the data
- ⊖ the method is not working well in HD spaces ($p > 50$)
 - The method is limited to continuous data

k-nearest neighbours (kNN):

k-NN is a very basic but still reference algorithm for classification. The spirit of k-NN is to compare any new observation x^* to the learning data and classify it according to its neighbors.

Algorithm: input x^* , k

- (i) find in the learning data set (X, Y) the k nearest neighbors of x^* Learning
- (ii) assign x^* to the class which is the most represented among the neighbors. Prediction



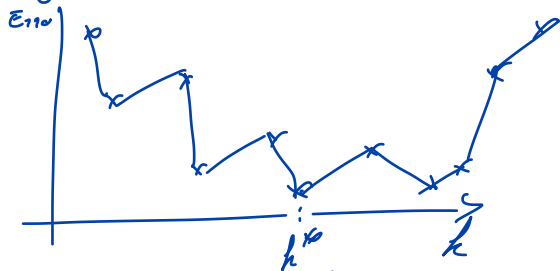
$$k = 4$$

$$y_{k-NN}^* = \text{"blue"}$$

Remark: the choice of k in $k-NN$ is a very sensitive question since it may end up with different classifications for different values of k .

$\Rightarrow$ the appropriate way to choose k is to rely on cross-validation.

Prob: even with CV, the choice of h may be quite difficult because the curve of the classif. error according to h will be quite irregular



Prob: h -NN can be applied to any types of data if we are able to define a distance

- continuous data:
 - euclidean
 - Manhattan

- graph:
 - shortest path
 - average random walk

- text:
 - ...

Summary of k-NN:

- ⊕ simple and easy to understand method
- ⊕ it can be applied to different data types
- ⊖ the choice of k is sensitive and not easy even with CV.
- ⊖ need to keep the whole learning data set and to find the neighbors in this large data set (which may be large!).